



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

2010
Year 11 Yearly
Examination

Mathematics

(2 Unit)

General Instructions

- Reading Time – 5 Minutes
- Working time – 90 Minutes
- Write using black or blue pen. Pencil may be used for diagrams.
- Board approved calculators maybe used.
- All necessary working should be shown in every question.
- All answers to be given in simplified exact form unless otherwise stated.
- Hand in your answers in 2 separate bundles:

Section A (Question 1, Question 2 and Question 3), and

Section B (Question 4, Question 5 and Question 6)

Total Marks – 70

- Attempt questions 1-6
- All questions are **NOT** of equal value.

Examiner: *P. Bigelow*

Section A – Start a new booklet.

Question 1 (10 marks).	Marks
a) The point $(2,c)$ lies on $x + 2y + 4 = 0$, find the value of c .	2
b) Find T_{10} of $5 + 9 + 13 + 17 + \dots$	1
c) If $f(y) = 9 - y^2$, find: (i) $f(-2)$ (ii) $f(y+1)$	2
d) Solve $x^2 + 2x - 8 = 0$.	1
e) State the domain of $f(x) = \sqrt{3-x}$.	1
f) Find x in the following: (i) $\log_x 36 = 2$ (ii) $\log_8 128 = x$	2
g) Form a quadratic equation with roots 2 and -5.	1

End of Question 1

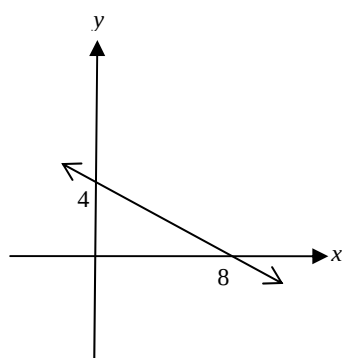
Question 2 (10 Marks).**Marks**

a) Find the sum of the first 9 terms of $2 - 1 + \frac{1}{2} - \frac{1}{4} + \dots$

1

b) Write down the equation of the line represented in the diagram:

2



c) State whether the following functions are ODD, EVEN or NEITHER:

3

(i) $f(x) = \frac{4}{9 + x^2}$

(ii) $f(x) = \frac{x}{9 + x}$

(iii) $f(x) = \frac{-x}{9 + x^2}$

d) Solve $2^{3x+2} = 64$

2

e) Use the quadratic formula to solve $2x^2 + 5x - 1 = 0$. (Leave answers in surd form.)

2

End of Question 2

Question 3 (11 marks).**Marks**

- a) By considering $0.\dot{5}\dot{4}$ as an infinite geometric series, express $0.\dot{5}\dot{4}$ as a fraction in simplest form. 2
- b) Sketch the following lines on separate diagrams (showing essential features): 4
- (i) $y = x + 2$
- (ii) $3x - y - 6 = 0$
- c) State whether the following quadratics are INDEFINITE, POSITIVE DEFINITE or NEGATIVE DEFINITE: 3
- (i) $2x^2 + 3x + 7$
- (ii) $6 - x - x^2$
- (iii) $x^2 - 9x - 8$
- d) Find $\lim_{x \rightarrow 4} \frac{x-4}{x^2-16}$. 2

End of Question 3**End of Section A**

Section B – Start a new booklet.**Question 4 (13 marks).****Marks****a)** Sketch the following on separate diagrams (showing essential features):**6**

(i) $xy = -4$

(ii) $y = 2^{-x}$

(iii) $y = \sqrt{4 - x^2}$

(iv) $x^2 + y^2 - 6y = 0$

b) If α and β are the roots of $2x^2 - 6x - 1 = 0$, find:**4**

(i) $\alpha + \beta$

(ii) $\alpha\beta$

(iii) $\frac{1}{\alpha} + \frac{1}{\beta}$

(iv) $\alpha^2 + \beta^2$

c) If $f(x) = x^2 + 2x + 5$, find $f'(x)$ where $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$.**3****End of Question 4**

Question 5 (14 marks).**Marks**

a) Graph the region defined by the intersection of $y \geq x^2$ and $x + y \leq 2$.

2

b) Find $f'(x)$ in the following:

6

(i) $f(x) = 3x^2 - x + 1$

(ii) $f(x) = (1 - 5x)^6$

(iii) $f(x) = \frac{1}{\sqrt{x}}$

c) If $\log_a 3 = 0.477$ and $\log_a 2 = 0.301$. Find:

6

(i) $\log_a 6$

(ii) $\log_a 9$

(iii) $\log_a 1.5$

End of Question 5

Question 6 (14 marks).**Marks**

- a)** (i) Find the axis of symmetry of $y = 4 + x - x^2$. 2
- (ii) Hence, or otherwise, find the maximum value of $4 + x - x^2$. 2
- b)** Find the equation of the two tangents to the curve $y = 3x^2 - 6x$ at the points where it crosses the x -axis. 4
- c)** Given $5^x = 13$, find x correct to two decimal places. 2
- d)** A couple wish to save for a deposit on a home. They need to save \$20,000 over a 5 year period. They deposit \$ P every month into an account which is paying 9%p.a., compounding monthly. 4
- (i) Show that $20000 = P(1.0075 + 1.0075^2 + \dots + 1.0075^{60})$.
- (ii) Find P to the nearest dollar.

End of Question 6**End of Section B.****End of Examination.**

Solutions YR 11 Unit Yearly 2010.

① (a) $x + 2y + 4 = 0$

$$\left. \begin{array}{l} x = 2 \\ y = c \end{array} \right\} \begin{array}{l} 2 + 2c + 4 = 0 \\ 2c + 6 = 0 \\ c = -3 \end{array} \quad (2)$$

(b) $5 + 9 + 13 + 17 + \dots$

Find T_{10} .
arithmetic

$$\begin{aligned} T_n &= a + (n-1)d \\ T_{10} &= 5 + 9 \times 4 \\ &= 41 \end{aligned} \quad (1)$$

(c) $f(y) = 9 - y^2$

(i) $f(-2) = 9 - (-2)^2 = 9 - 4 = 5 \quad (1)$

(ii) $f(y+1) = 9 - (y+1)^2 = 9 - (y^2 + 2y + 1) = 9 - y^2 - 2y - 1$
 $= 8 - y^2 - 2y \quad (1)$

(d) $x^2 + 2x - 8 = 0$

$(x+4)(x-2) = 0$

$x = -4, x = 2. \quad (1)$

(e) $f(x) = \sqrt{3-x}$

$3-x \geq 0$

$-x \geq -3 \quad (1)$

$x \leq 3$

(f) (i) $x^2 = 36 \quad x = 6 \quad (1)$

(ii) $8^x = 128$

$2^{3x} = 2^7 \Rightarrow 3x = 7 \quad x = \frac{7}{3} \text{ or } 2\frac{1}{3} \quad (1)$

Q 1 (a) $(x-2)(x+5) = 0$
 or $x^2 + 5x - 2x - 10 = 0$
 $x^2 + 3x - 10 = 0$

(1)

Q 2 (a) $2 - 1 + \frac{1}{2} - \frac{1}{4} + \dots$

geometric

$a = 2, n = 9, r = -\frac{1}{2}$

$$S_n = \frac{a(r^n - 1)}{(r - 1)} = \frac{2 \left(\left(-\frac{1}{2}\right)^9 - 1 \right)}{-\frac{1}{2} - 1} = 2 \times \frac{2}{3} \left(\left(-\frac{1}{2}\right)^9 - 1 \right)$$

$$= -\frac{4}{3} \left(-\frac{1}{512} - 1 \right) = 1 \frac{43}{128} \quad (1)$$

(b) points are $(8, 0)$ and $(0, 4)$

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{4 - 0}{0 - 8} = -\frac{1}{2}$$

use $(y - y_1) = m(x - x_1)$ and $(8, 0)$

$$(y - 0) = -\frac{1}{2}(x - 8)$$

$$2y = -x + 8$$

$$x + 2y - 8 = 0$$

(2)

(c) (i) $f(x) = \frac{4}{9+x^2}$ $f(-x) = \frac{4}{9+(-x)^2} = \frac{4}{9+x^2}$ even (1)

(ii) $f(x) = \frac{x}{9+x}$ $f(-x) = \frac{-x}{9-x} = \frac{-x}{-(x-9)} = \frac{x}{x-9}$
 neither (1)

$$2 \text{ (c) (iii) } f(x) = \frac{-x}{9+x^2}$$

$$f(-x) = \frac{-(-x)}{9+(-x)^2} = \frac{x}{9+x^2}$$

$$\text{odd is } f(x) = -f(-x) \text{ so } -\left(\frac{x}{9+x^2}\right) = \frac{-x}{9+x^2}$$

odd. (1)

$$(d) \quad 2^{3x+2} = 64$$

$$2^{3x+2} = 2^6$$

$$3x+2 = 6$$

$$3x = 4 \quad x = \frac{4}{3} \quad (2)$$

$$(e) \quad 2x^2 + 5x - 1 = 0$$

$$x = \frac{-5 \pm \sqrt{25 - 4 \times 2 \times -1}}{4}$$

$$= \frac{-5 \pm \sqrt{33}}{4} \quad (2)$$

$$Q3. (a) 0.\dot{5}4 = 0.54 + 0.0054 + 0.000054 + \dots$$

$$\text{geometric } a = 0.54$$

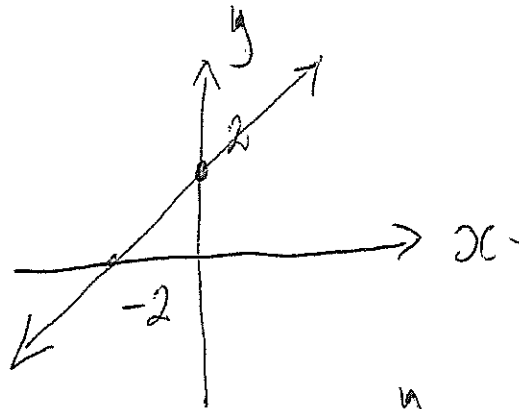
$$r = 0.01$$

$$S_{\infty} = \frac{a}{1-r} = \frac{0.54}{1-0.01}$$

$$= \frac{0.54}{0.99} = \frac{54}{99}$$

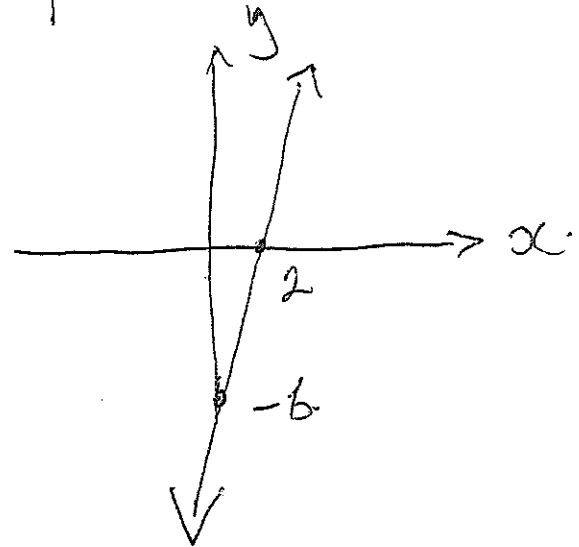
$$(2) \quad = \frac{6}{11}$$

3 (b) (i) $y = x + 2$.



(2)

(ii) $3x - y - 6 = 0$
 $y = 3x - 6$



(2)

(c) $\Delta = b^2 - 4ac$

(i) $2x^2 + 3x + 7$ $\Delta = 9 - 4 \times 2 \times 7 = -47$

concave up. So positive def (1)
 no roots

(ii) $-x^2 - x + 6$ $\Delta = 1 - 4 \times (-1) \times 6$
 $= 25$

concave down indefinite (1)
 has roots

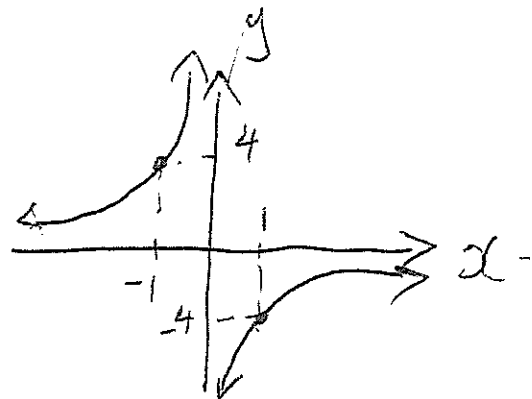
(iii) $x^2 - 9x - 8$ $\Delta = 81 - 4 \times 1 \times (-8)$
 $= 113$

concave up indefinite (1)
 has roots

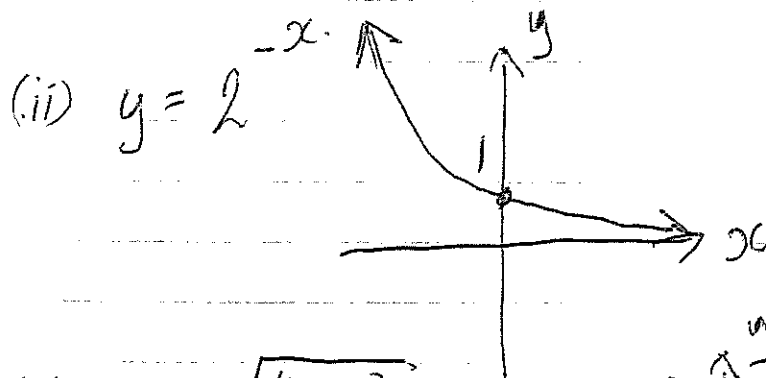
(d) $\lim_{x \rightarrow 4} \frac{(x-4)}{(x-4)(x+4)} = \frac{1}{x+4} \rightarrow \frac{1}{8}$ (2)

Q4

(a) (i) $xy = -4$
 $y = -\frac{4}{x}$

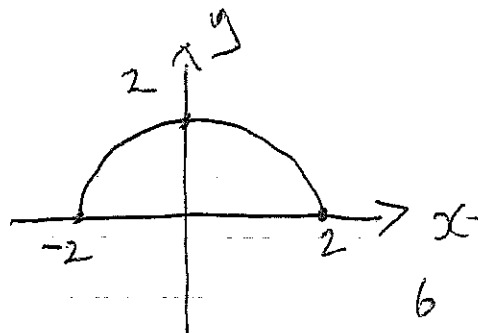


(1)



(1)

(iii) $y = \sqrt{4-x^2}$



(2)

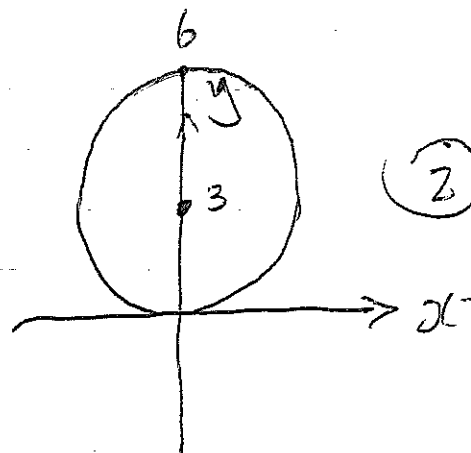
$(-\frac{6}{2})^2$

(iv) $x^2 + y^2 - 6y = 0$

$x^2 + y^2 - 6y + 9 = 9$

$x^2 + (y-3)^2 = 9$

Circle centre (0,3) r=3.



(2)

(b) $2x^2 - 6x - 1 = 0$

(i) $\alpha + \beta = -\frac{b}{a} = -\frac{-6}{2} = 3$ (1)

(ii) $\alpha\beta = \frac{c}{a} = -\frac{1}{2}$ (1)

(iii) $\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta} = \frac{3}{-\frac{1}{2}} = -6$ (1)

(iv) $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = 3^2 - 2 \times -\frac{1}{2} = 9 + 1 = 10$ (1)

4 (c) $f(x) = x^2 + 2x + 5$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{(x+h)^2 + 2(x+h) + 5 - (x^2 + 2x + 5)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\cancel{x^2} + 2xh + \cancel{h^2} + \cancel{2x} + 2h + 5 - \cancel{x^2} - \cancel{2x} - 5}{h}$$

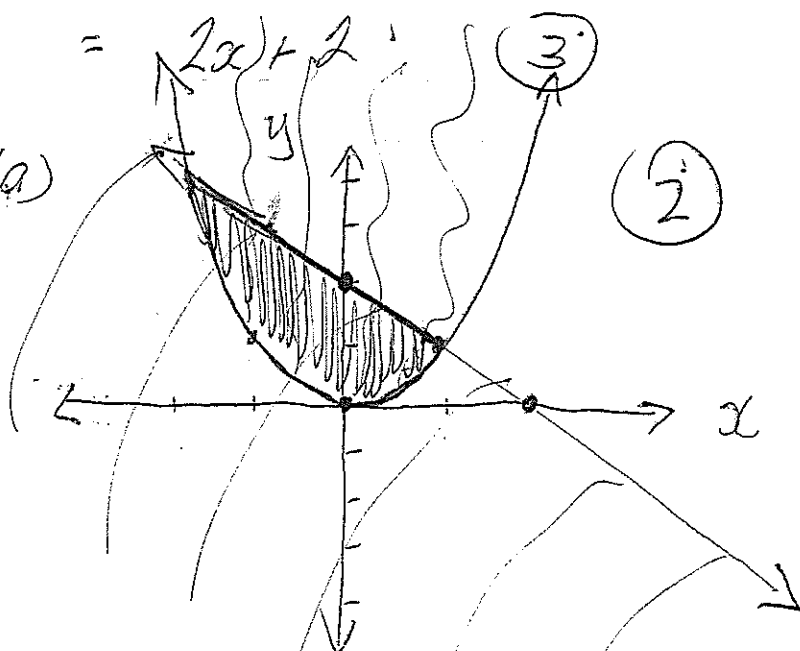
$$= \lim_{h \rightarrow 0} \frac{2xh + h^2 + 2h}{h}$$

$$= \lim_{h \rightarrow 0} \cancel{h} (2x + h + 2)$$

$$= 2x + 2$$

(P5)

(a)



(2)

$$y \geq x^2 \text{ and } x + y \leq 2.$$

(b) (i) $f'(x) = 6x - 1$

(2)

(ii) $f'(x) = 6(1-5x)^5 \cdot x - 5 = -30(1-5x)^5$ (2)

(iii) $f(x) = x^{-\frac{1}{2}}$, $f'(x) = -\frac{1}{2}x^{-\frac{1}{2}} = \frac{-1}{2x\sqrt{x}}$ (2)

$$\begin{aligned}
 \text{Q 5 (c) (i) } \log_a 6 &= \log_a (3 \times 2) \\
 &= \log_a 3 + \log_a 2 \\
 &= 0.477 + 0.301 \\
 &= 0.778 \quad (2)
 \end{aligned}$$

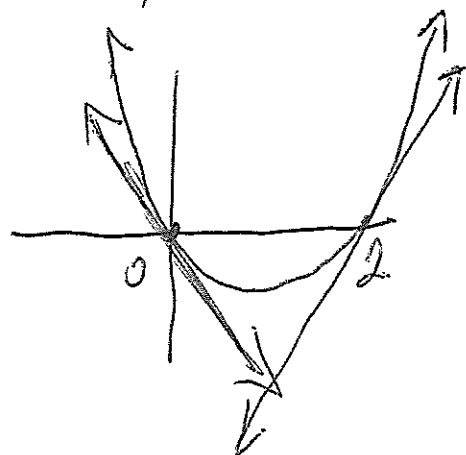
$$\begin{aligned}
 \text{(ii) } \log_a 9 &= \log_a 3^2 \\
 &= 2 \log_a 3 = 2 \times 0.477 \\
 &= 0.954 \quad (2)
 \end{aligned}$$

$$\begin{aligned}
 \text{(iii) } \log_a \frac{3}{2} &= \log_a 3 - \log_a 2 \\
 &= 0.477 - 0.301 \\
 &= 0.176 \quad (2)
 \end{aligned}$$

$$\begin{aligned}
 \text{Q 6. (a) (i) } y &= -x^2 + x + 4 \\
 \text{axis } x &= -\frac{b}{2a} = -\frac{1}{2 \times -1} = \frac{1}{2} \quad (2)
 \end{aligned}$$

$$\begin{aligned}
 \text{max } y \text{ value is } x = \frac{1}{2} \text{ into } y &= -x^2 + x + 4 \\
 &= -\left(\frac{1}{2}\right)^2 + \frac{1}{2} + 4 \\
 &= 4\frac{1}{4} \quad (2)
 \end{aligned}$$

$$\begin{aligned}
 \text{(b) } y &= 3x^2 - 6x \\
 &= 3x(x-2) \\
 \text{when } 3x(x-2) &= 0 \\
 x &= 0 \text{ and } x = 2.
 \end{aligned}$$



6 (b) cont

$$y = 3x^2 - 6x$$

$$y' = 6x - 6$$

$$\text{at } x=0 \quad m = 0 - 6 = -6$$

$$\text{at } x=2 \quad m = 6 \times 2 - 6 = 6$$

$$\text{So } (0,0) \quad m = -6$$

$$(y-0) = -6(x-0)$$

$$y = -6x \quad (2)$$

$$\text{and } (2,0) \quad m = 6$$

$$(y-0) = 6(x-2)$$

$$y = 6x - 12 \quad (2)$$

$$(c) \quad 5^x = 13$$

$$\log_{10} 5^x = \log_{10} 13$$

$$x \log_{10} 5 = \log_{10} 13$$

$$x = \frac{\log_{10} 13}{\log_{10} 5}$$

$$\approx 1.59$$

(2)

(d)

(6) (d) \$20 000
 5 years -
 \$P every month.
 9% p.a. $\Rightarrow$ 0.75% per month.

1st month $P + 0.75\% P = P(1 + 0.75\%)^{60}$

2nd month $P(1.0075)^{59} = P(1.0075)^{60}$

Last month $P(1.0075)^1$

So $P(1.0075)^1 + P(1.0075)^2 + \dots + P(1.0075)^{60} = 20000$

So $P = \frac{20,000}{1.0075 + 1.0075^2 + \dots + 1.0075^{60}}$

denominator $\frac{r^b - a}{r - 1} = \frac{1.0075 \times 1.0075^{60} - 1.0075}{1.0075 - 1}$

$= \frac{1.0075^{61} - 1.0075}{0.0075}$

$= 75.98981795 \dots$

$P = \$263.19$
 $\approx \$263.$

(2)